1. Morphodynamic Feedback Architecture & Master Equation Matrix

Physical Foundation: Why Sand Dunes Are Self-Organizing Aerodynamic-Granular Waves

A sand dune is not a static pile of grains; it is a dynamic wave emerging from the closed feedback loop between a turbulent fluid boundary layer (wind or water) and a mobile granular bed:

(1) Topography h(x, y, t) perturbs the turbulent boundary layer, accelerating flow over the stoss slope and detaching into a recirculating wake eddy behind steep lee slopes (> 14 deg).

(2) Because of fluid inertia in the inner shear layer, the basal shear stress maximum tau_max is shifted UPWIND of the topographic crest (Jackson & Hunt 1975). Conversely, grain inertia causes the actual sand flux q(x) to lag DOWNWIND of the shear stress by the saturation length L_sat (Claudin & Andreotti 2006).

(3) The competition between the aerodynamic upwind phase advance and the inertial downwind saturation lag determines whether bed perturbations amplify into dunes (lambda > lambda_min) or decay into a flat bed.

Master Guide: Core Equations, When Each Is Triggered & Why It Is Essential

1. Drag & Saturation Scales (constants.py): L_drag = (rho_s / rho_f) d, L_sat = 2.2 L_drag
   - WHEN USED: Initialization of any planetary environment and at every spatial flux relaxation step.
   - WHY USED: Sets the distance required for saltating grains to accelerate to fluid speed; controls the minimum and fastest-growing dune wavelengths across Earth (0.95 m), Mars (44 m), Venus (1.4 cm), and water (1.2 mm).

2. Separation Bubble Envelope (separation.py): h_tilde(x) = max(h(x), s_Hermite(x)) when dh/dx < -tan(14 deg)
   - WHEN USED: Step 1 of every time step whenever any lee slope exceeds the critical detachment angle (14 deg).
   - WHY USED: Turbulent boundary layers cannot remain attached down a steep slipface. The cubic Hermite streamline encloses the zero-shear wake eddy (L_b ~ 6 H_d) and provides a smooth C^1 envelope for the FFT wind solver.

3. Jackson-Hunt / Kroy FFT Shear Stress (wind.py): tau_pert_hat(k) = [A (kx^2/|k|) + i B kx] h_tilde_hat(k)
   - WHEN USED: Step 2 of every time step over the aerodynamic envelope h_tilde(x, y); masked to 0 inside wake.
   - WHY USED: Captures nonlocal Bernoulli pressure speed-up over the crest (A |k|) and the inner-layer upwind phase shift (+i B kx) that drives sand erosion on the windward stoss and deposition near the crest.

4. Saturated Flux & Slope Threshold (transport.py): q_sat = C (rho_f/g) u_*^3 (1 - tau_th,eff / tau)_+
   - WHEN USED: Step 3a of every time step where basal shear tau exceeds slope-adjusted threshold tau_th,eff.
   - WHY USED: Quantifies the maximum steady-state mass transport capacity of saltation, while the Bagnold-Howard gravity correction tau_th,eff = tau_th (1 + dh/dx / tan 34 deg) prevents unphysical stoss over-steepening.

5. Non-Local Exponential Relaxation (transport.py): L_sat dq_x/dx = q_sat - q_x & q_y = q_x(tau_y/tau_0 - D_lat dh/dy)
   - WHEN USED: Step 3b of every time step, sweeping downwind in +x and across transverse faces j + 1/2 in y.
   - WHY USED: Solves the spatial lag of saltation adaptation (essential to damp short-wavelength noise) and models lateral wind deflection + splash diffusion in 2D, while capping entrainment on bare non-erodible bedrock.

6. Exner Conservation & Isotropic Avalanche (exner.py, avalanche.py): rho_b dh/dt + div(q) = 0, |grad h| <= tan(32 deg)
   - WHEN USED: Steps 4-5 of every time step (upwind finite-volume step + Jacobi slope relaxation when |grad h| > tan 34 deg).
   - WHY USED: Strictly conserves total sand volume to machine precision (< 1e-13) and relaxes over-steepened lee faces isotropically to the dynamic angle of repose (32 deg), forming the razor-sharp Barchan slipface.

Closed-Loop Execution Pipeline in Every Time Step dt (DuneSimulator1D & DuneSimulator2D)

$$h(x,y,t) \longrightarrow \tilde{h}(x,y), \text{ wake} \longrightarrow \tau_x, \tau_y \longrightarrow q_x, q_y \longrightarrow h^*(x,y) \longrightarrow h(x,y,t+\Delta t)$$

Key Numerical Guarantee: Every spatial operator is matched to the physical characteristic direction of information propagation: pseudo-spectral nonlocal FFT for elliptic pressure-shear perturbation, exact exponential integration for hyperbolic downwind flux relaxation, donor-cell upwinding for Exner divergence, and flux-conservative isotropic edge relaxation for granular avalanching.

2. Planetary Boundary Layer Parameters & Universal Length Scales

Governing Equations in core/constants.py (Bagnold 1941; Claudin & Andreotti 2006)

Given gravitational acceleration g [m/s^2], fluid density rho_f [kg/m^3], grain density rho_s [kg/m^3], mean grain diameter d [m], and granular bed packing fraction phi_p = 0.62:

$$\rho_{\text{bulk}} = \phi_p \rho_s, \qquad L_{\text{drag}} = \frac{\rho_s}{\rho_f} d, \qquad L_{\text{sat}} = 2.2\, L_{\text{drag}} = 2.2\, \frac{\rho_s}{\rho_f} d$$

$$u_{*\text{th}} = A_B \sqrt{\frac{\rho_s - \rho_f}{\rho_f} g\, d} \quad (A_B = 0.10), \qquad \tau_{\text{th}} = \rho_f u_{*\text{th}}^2, \qquad \tau_0 = \rho_f u_{*0}^2$$

$$\lambda_{\min} = 2\pi \frac{A}{B} L_{\text{sat}} = 4\pi L_{\text{sat}} \approx 12.57\, L_{\text{sat}}, \qquad \lambda_{\max} \approx 19.78\, L_{\text{sat}}$$

- WHEN USED: Evaluated when instantiating a PlanetaryEnvironment (EARTH_DESERT, MARS_DUNES, VENUS_DUNES, AQUEOUS_FLUME) and referenced at every transport, dispersion, and Exner conservation step.
- WHY USED (Physical Derivation):
  * A stationary grain of mass m ~ rho_s d^3 entrained into a fluid stream of density rho_f experiences an aerodynamic drag force F_D ~ rho_f u^2 d^2. By Newton's second law (m du/dt = m u du/dx ~ F_D), the distance required to accelerate the grain to the fluid speed is L_drag = (rho_s / rho_f) d — independent of wind speed and gravity! Because saltation saturation involves grain-bed splash cascades, wind-tunnel and field measurements (Claudin & Andreotti 2006) show L_sat = 2.2 L_drag.
  * Because L_sat scales inversely with fluid density rho_f, thin Martian air (rho_f = 0.015 kg/m^3) produces L_sat = 44.0 m and giant 858 m dunes, whereas dense water (rho_f = 1000 kg/m^3) produces L_sat = 1.17 mm and 2.3 cm current ripples — explaining five orders of magnitude of bedform sizes with a single law!

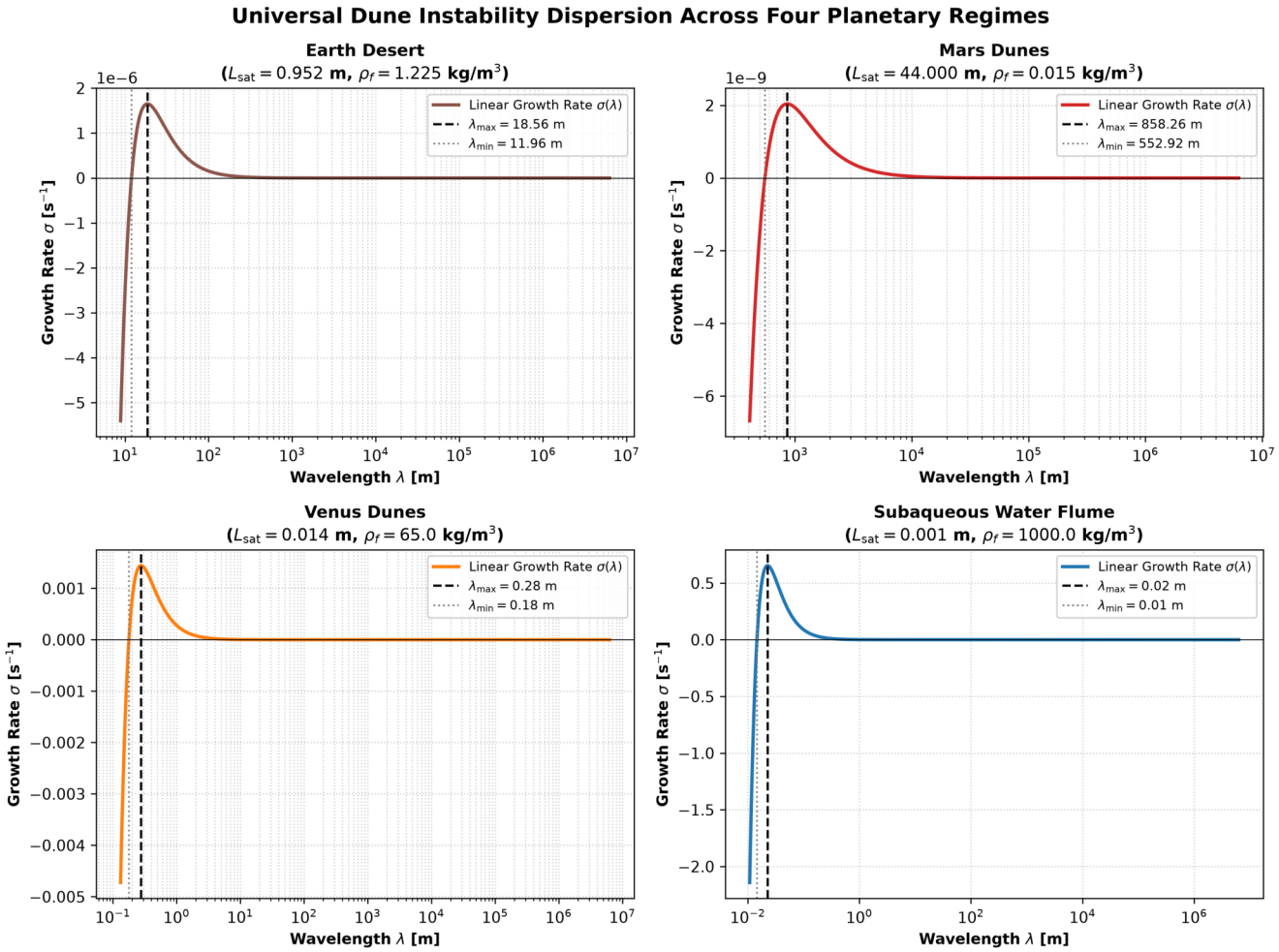


Figure 1 (from 05_planetary_comparison.py): Linear growth rate sigma(lambda) across Earth, Mars, Venus, and Subaqueous regimes.

3. Flow Separation Bubble & Aerodynamic Envelope Model

Governing Equations in core/separation.py (Kroy, Sauermann & Herrmann 2002)

Flow detachment occurs at any brink point x_d where the downward lee slope exceeds theta_sep = 14 deg:

$$\left.\frac{\partial h}{\partial x}\right|_{x_d^+} < -s_{\text{crit}} = -\tan(14^\circ) \approx -0.2493, \qquad H_d = h(x_d) - h_{\text{base}}, \qquad \bar{s}' = 0.25$$

With clamped upwind launch slope h'_d = clip(dh/dx(x_d^-), -s_bar', +s_bar') and dimensionless parameter nu = h'_d / s_bar', the separation bubble reattachment distance L_b and cubic Hermite streamline s(z) are:

$$L_b = \frac{1.5\,H_d}{\bar{s}'}\left(1 + \tfrac{1}{4}\nu + \tfrac{1}{8}\nu^2\right), \qquad z = \tfrac{x - x_d}{L_b} \in [0, 1]$$

$$s(z) = h_{\text{base}} + (2H_d + h'_d L_b)z^3 - (3H_d + 2h'_d L_b)z^2 + (h'_d L_b)z + H_d$$

$$\tilde{h}(x) = \max\left(h(x),\, s(x)\right), \qquad \text{in_wake}(x) = \mathbf{1}(s(x) > h(x))$$

- WHEN USED: Computed before the wind solver at every time step in 1D and along every streamwise slice in 2D.
- WHY USED (Physical & Mathematical Rationale):
  1. Adverse Pressure Gradient Detachment: When air flows over a sharp brink onto a 32-deg slipface, the rapid expansion creates a strong adverse pressure gradient that detaches the boundary layer, forming a recirculating Karman-like eddy extending ~6 H_d downwind.
  2. Exact Hermite Boundary Conditions: The cubic polynomial s(z) is the unique lowest-order curve satisfying all four physical boundary conditions of a dividing streamline: C^0 & C^1 launch continuity at the brink (s(0) = h(x_d), s'(0) = h'_d) and smooth tangential reattachment at the bed (s(1) = h_base, s'(1) = 0).
  3. Outer Flow Regularization: Passing the smooth envelope h_tilde(x) instead of the sharp-kinked bed h(x) to the FFT wind solver prevents Gibbs ringing, while setting tau = 0 inside in_wake forces all saltating grains crossing the brink to deposit immediately onto the upper slipface.

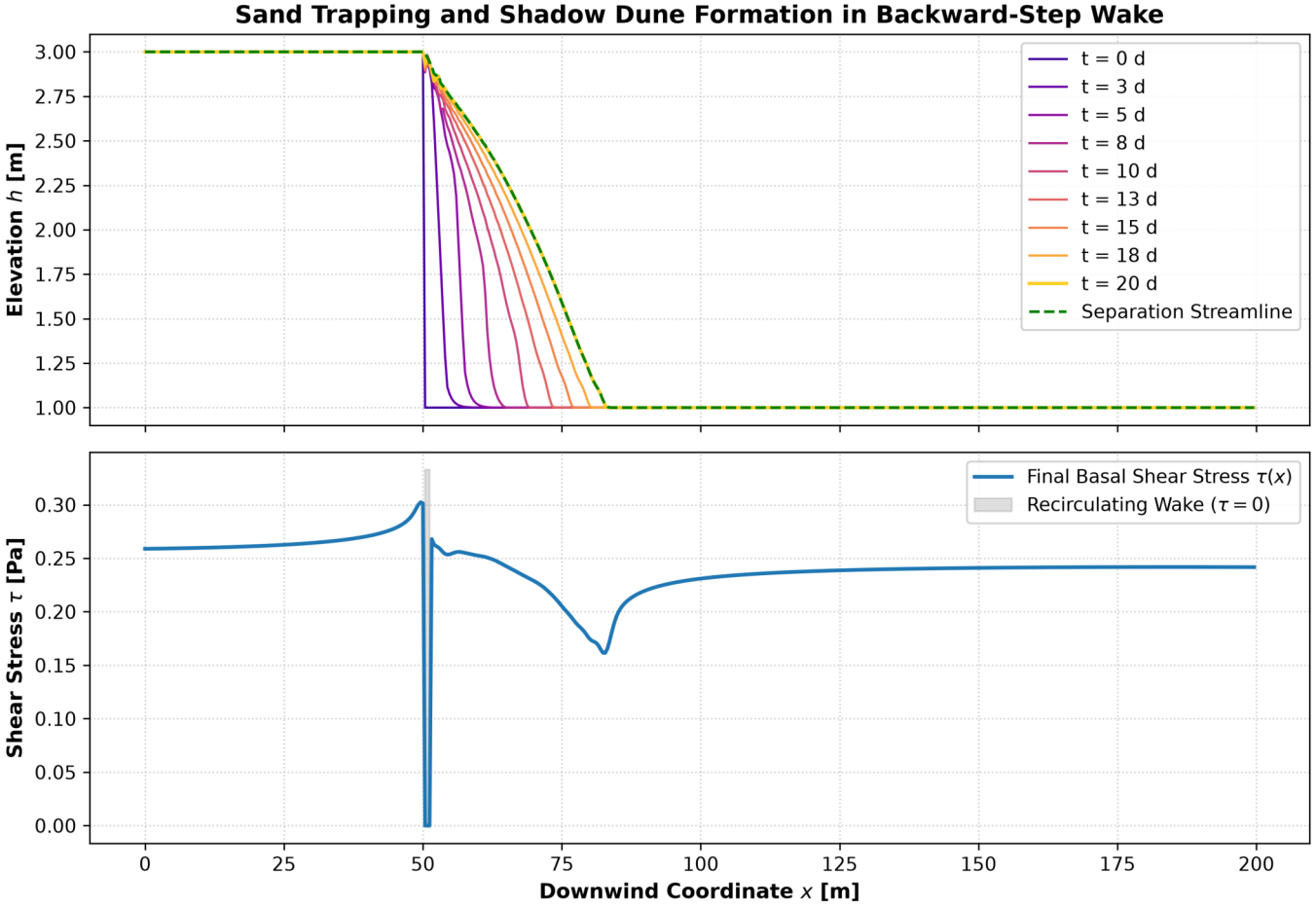


Figure 2 (from 03_step_topography_response.py): Flow detachment, L_b = 6 H_d wake bubble, and shadow dune trapping behind a step.

4. Jackson-Hunt FFT Shear Stress Solver & Primary Instability

Governing Equations in core/wind.py & analysis/dispersion.py (Jackson & Hunt 1975; Fourriere et al. 2010)

In Fourier space, the relative basal shear stress perturbation over the aerodynamic envelope h_tilde is:

$$\hat{\tau}_{\text{pert},x}(k_x,k_y) = \left(A\,\frac{k_x^2}{|k|} + i\,B\,k_x\right) e^{-\frac{1}{2}(|k|\ell_s)^2}\,\hat{\tilde{h}}(k_x,k_y), \qquad \hat{\tau}_{\text{pert},y}(k_x,k_y) = A_y\,\frac{k_x k_y}{|k|} e^{-\frac{1}{2}(|k|\ell_s)^2}\,\hat{\tilde{h}}(k_x,k_y)$$

In 1D spatial coordinates, this is equivalent to the nonlocal Hilbert transform convolution:

$$\tau_{\text{pert}}(x) = A\,\mathcal{H}\left[\frac{\partial \tilde{h}}{\partial x}\right](x) + B\,\frac{\partial \tilde{h}}{\partial x}(x), \qquad \tau(x) = \tau_0 \max\left(0,\, 1 + \tau_{\text{pert}}(x)\right) \;\; (\tau = 0 \text{ in wake})$$

Linearizing the coupled (tau, q, h) system around a flat bed yields the exact dispersion relation:

$$\sigma(k) = \frac{Q_1}{\rho_{\text{bulk}}}\,\frac{k^2\,(B - A\,|k|\,L_{\text{sat}})}{1 + (k\,L_{\text{sat}})^2}, \qquad c(k) = \frac{Q_1}{\rho_{\text{bulk}}}\,\frac{|k|\,(A + B\,|k|\,L_{\text{sat}})}{1 + (k\,L_{\text{sat}})^2}$$

- WHEN USED: Step 2 of every time step (FFT shear solver) and in spectral stability diagnostics.
- WHY USED (Physical Meaning of Every Term):
  * In-Phase Term A |k| (A = 3.2): Potential outer-flow streamline compression (Bernoulli effect). It is symmetric and peaks at the crest. In the dispersion relation sigma(k), it appears as -A |k| L_sat (stabilizing short waves!).
  * Quadrature Term +i B k (B = 1.6): Turbulent shear-stress phase advance in the inner layer (thickness l). It is proportional to the upwind slope dh/dx, shifting the peak shear stress UPWIND of the crest by angle arctan(B/A). In sigma(k), +B is the SOLE positive destabilizing term that pumps sand from troughs to crests!
  * Transverse Term A_y kx ky / |k| (A_y = 1.6): Lateral pressure gradient deflecting wind around 3D mounds.

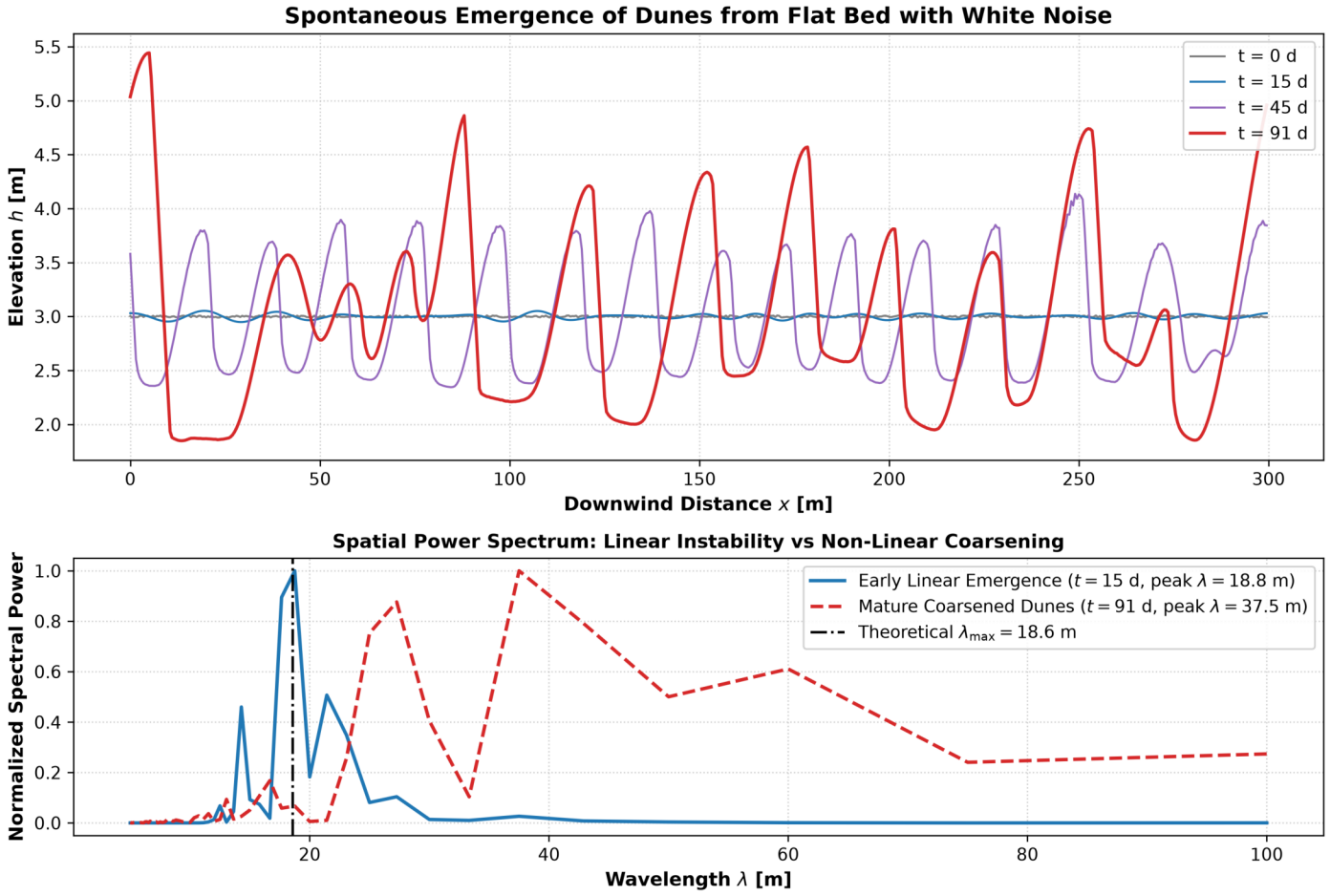


Figure 3 (from 01_spontaneous_dune_formation.py): Spontaneous emergence of periodic dunes at lambda_max ~= 19.8 L_sat from 1 cm noise.

5. Non-Local Saltation Flux, Slope Threshold & Bedrock Limiting

Governing Equations in core/transport.py (Sauermann, Kroy & Herrmann 2001; Duran et al. 2010)

1. Slope-Modified Threshold & Saturated Sand Flux Capacity q_sat [kg/(m s)]:

$$\tau_{\mathrm{th,\,eff}}(x,y) = \tau_{\mathrm{th},0}\,\mathrm{clip}\Big(1 + \tfrac{\partial_x h}{\tan\theta_{\mathrm{rep}}},\ 0.2,\ 2.5\Big), \qquad q_{\mathrm{sat}}(\tau) = C_{\mathrm{flux}}\,\tfrac{\rho_f}{g}\,u_*^3\,\big(1 - \tfrac{\tau_{\mathrm{th,\,eff}}}{\tau}\big)_+$$

2. Non-Local Spatial Relaxation PDE & Exact Exponential Euler Recurrence (with mu = exp(-dx / L_sat)):

$$L_{\mathrm{sat}}\,\tfrac{\partial q_x}{\partial x} = q_{\mathrm{sat}}(x) - q_x(x) \quad\Longrightarrow\quad q_{x,i}^{\mathrm{ideal}} = q_{x,i-1}\,\mu + q_{\mathrm{sat},i}\,(1-\mu)$$

3. Non-Erodible Bedrock Limiter & Donor-Cell Transverse Flux across y-interfaces j + 1/2:

$$q_{x,i} = \min\big(q_{x,i}^{\mathrm{ideal}},\ q_{x,i-1} + \tfrac{\rho_{\mathrm{bulk}}\,\Delta x}{\Delta t}\max(0, h_i - h_{\mathrm{floor}})\big), \qquad q_{y,j+1/2} = q_{x,\mathrm{up}}\,\tfrac{\tau_y}{\tau_0} - D_{\mathrm{lat}}\,q_{x,\mathrm{donor}}\,\tfrac{h_{j+1}-h_j}{\Delta y}$$

- WHEN USED: Step 3 of every time step after computing basal shear stress tau.
- WHY USED (Physical & Numerical Necessity):
  1. Why Instantaneous Flux (q = q_sat) Fails Catastrophically: If L_sat -> 0, the growth rate becomes sigma(k) ~ B k^2, which grows fastest at k -> infinity (grid-scale 2*dx noise explodes in a few steps!). The spatial relaxation ODE L_sat dq/dx = q_sat - q introduces the physical phase delay that makes sigma(k) < 0 for all wavelengths shorter than lambda_min = 4*pi*L_sat.
  2. Why Exponential Euler Is Used: When dx >= L_sat (e.g., subaqueous ripples or coarse grids), standard forward Euler (1 - dx/L_sat) oscillates and goes negative. Exponential Euler mu = exp(-dx/L_sat) is unconditionally positive, monotone, and exact to machine precision (< 1e-12) for piecewise-constant q_sat.
  3. Why Forward Slope dh/dx Is Used in tau_th,eff: Pairing forward difference dh/dx in q_sat with backward difference dq/dx in the Exner equation forms a centered diffusive operator (h_{i+1} - 2h_i + h_{i-1})/dx^2 that damps Nyquist modes while naturally regulating the windward stoss angle to ~10 deg.

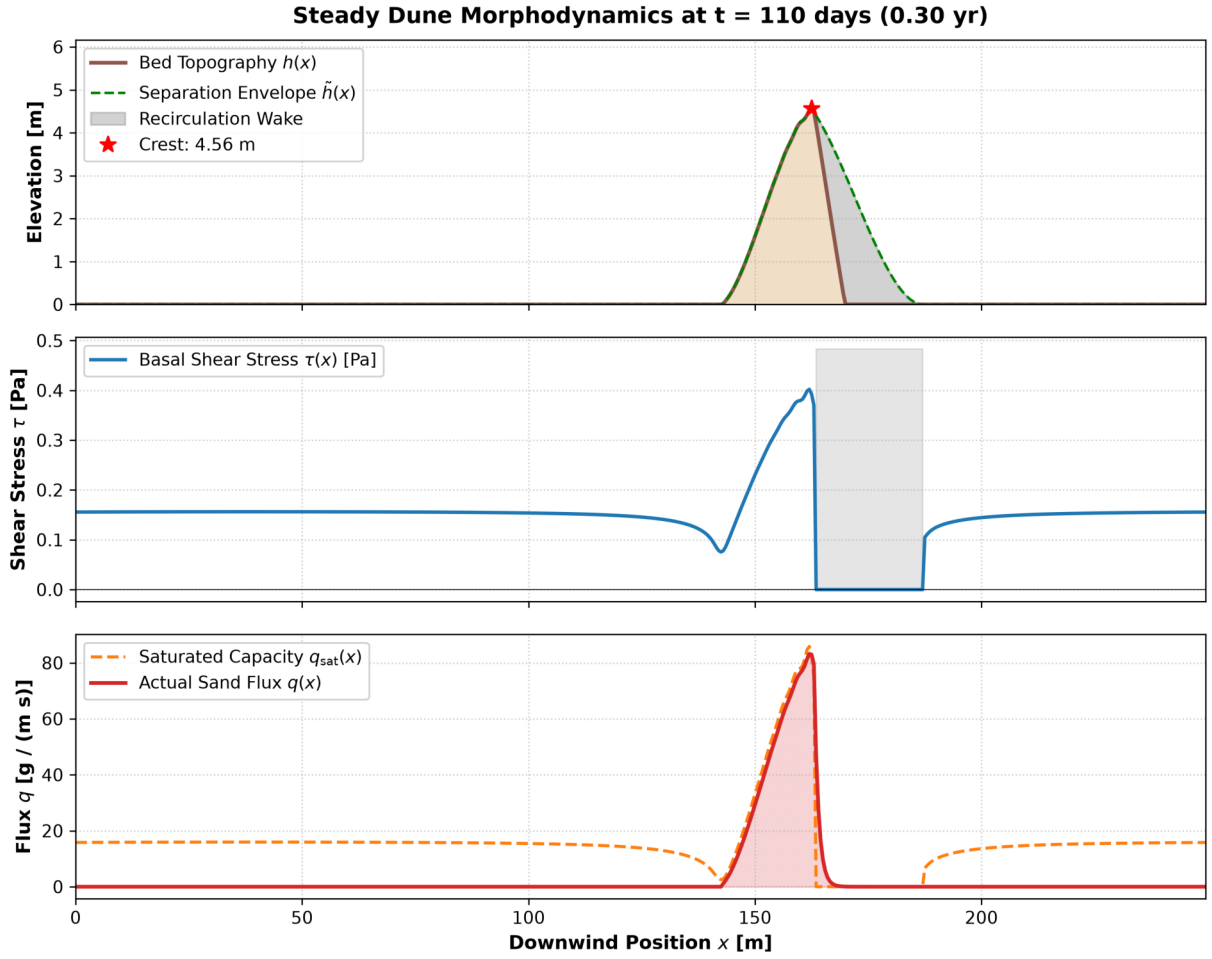


Figure 4 (from 02_gaussian_heap_evolution.py): Synchronized 3-panel profile of h(x), h_tilde(x), tau(x), q_sat(x), and relaxed q(x).

6. Exner Mass Conservation, Isotropic Avalanche & 1/H Migration Law

Governing Equations in core/exner.py & core/avalanche.py (Exner 1925; Bouchaud et al. 1994)

1. Upwind Finite-Volume Exner Equation & Morphological CFL Time Step:

$$\frac{\partial h}{\partial t} = -\frac{1}{\rho_{\text{bulk}}}\left(\frac{q_{x,i,j} - q_{x,i-1,j}}{\Delta x} + \frac{q_{y,i,j+1/2} - q_{y,i,j-1/2}}{\Delta y}\right), \qquad \Delta t = \text{clip}\left(\frac{\text{CFL}\,\Delta x \tan 32^\circ}{\max|\partial_t h|},\ \Delta t_{\min},\ \Delta t_{\max}\right)$$

2. Isotropic 2D Cellular Avalanche Relaxation (Triggered where Euclidean slope |grad h| > tan 34 deg):

$$|\nabla h|_e = \sqrt{\left(\frac{\partial h}{\partial x}\right)_e^2 + \left(\frac{\partial h}{\partial y}\right)_e^2}, \qquad F_{\text{aval},e} = \omega\,\Delta h_e \max\left(0,\ 1 - \frac{\tan 32^\circ}{|\nabla h|_e}\right)$$

3. Kinematic Dune Migration Speed & Inverse-Height Scaling Law:

$$c = \frac{q_{\text{brink}}}{\rho_{\text{bulk}} H} \propto \frac{1}{H}$$

- WHEN USED: Steps 4 and 5 of every time step.
- WHY USED (Physical & Numerical Rationale):
  1. Upwind Differencing in +x: Because sand transport is a hyperbolic wave propagating downwind (c > 0), central differencing of dq_x/dx is unconditionally unstable (dispersive ringing). First-order backward (upwind) differencing is monotone, telescoping (sum(div_q) = 0 to < 1e-14), and shock-capturing.
  2. Why Isotropic 2D Euclidean Slope |grad h| Is Critical for Barchan Dunes: A Barchan slipface curves from 0 deg at the centerline to ~45 deg along the inner horns. If an avalanche solver checks |dh/dx| > tan 34 deg and |dh/dy| > tan 34 deg separately along grid axes, diagonal faces at 45 deg can stand at tan(32 deg)/cos(45 deg) = tan(41.5 deg) — 30% steeper than the center, creating false twin peaks on the shoulders! Using the Euclidean norm |grad h|_e relaxes all slipface orientations isotropically to 32 deg.
  3. Why c ~ 1/H Forms Barchan Horns: Integrating dh/dt = -c dh/dx = -(1/rho_b) dq/dx across the slipface [x_b, x_t] where q(x_t) = 0 inside the wake gives c H = q_brink / rho_b. Because the lateral flanks of a 3D mound have smaller height H(y) < H_center, they migrate faster (c(y) ~ 1/H(y)), advancing ahead to form the two horns!

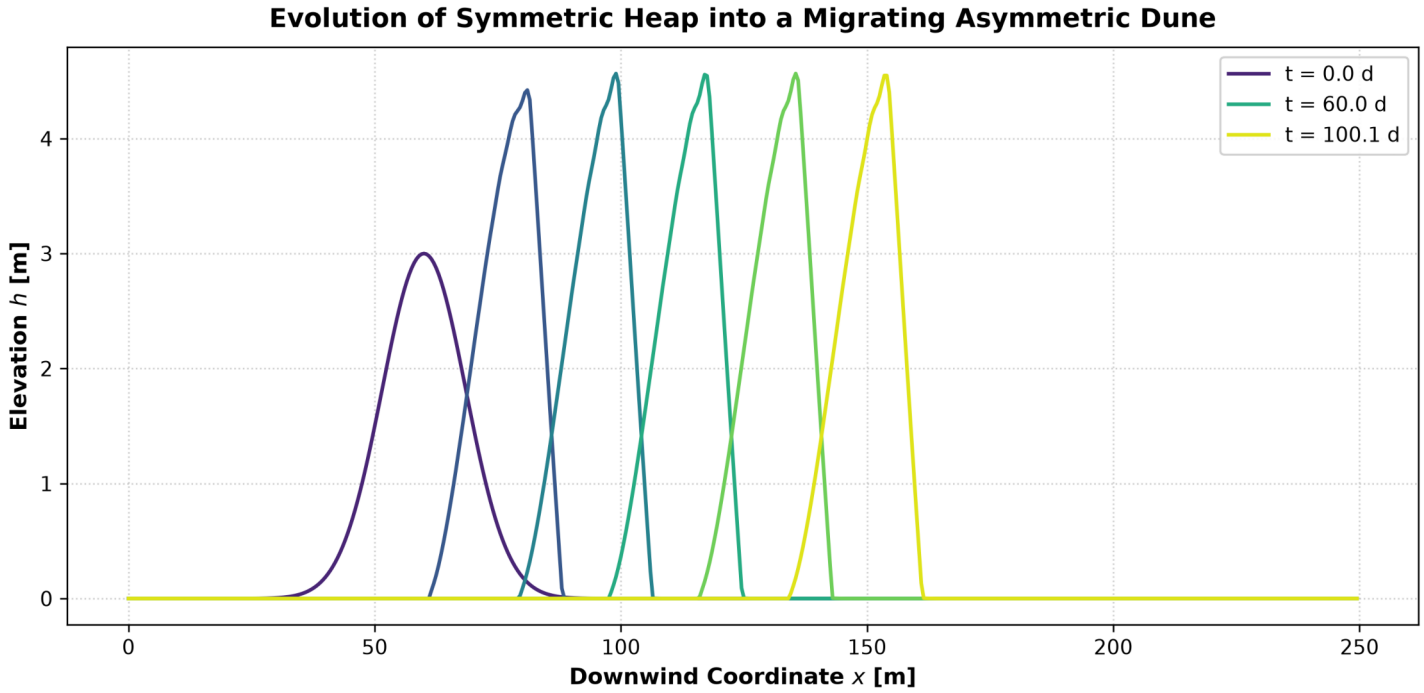


Figure 5 (from 02_gaussian_heap_evolution.py): Transformation of a symmetric Gaussian heap into a shape-invariant migrating dune.

7. Synthesis: Emergent 2D Crescentic Barchan Dune Morphodynamics

How the Five Physical Equations Jointly Sculpt a Crescentic Barchan Dune from a Circular Mound

Starting from a radially symmetric parabolic sand dome on flat, non-erodible bedrock (t = 0 d):

- Stage 1 (0 - 15 d, Stoss Flattening & Slipface Birth): Upwind shear shift (+i B kx) erodes the windward toe (modulated by gravity threshold tau_th,eff) and deposits near the crest until the lee slope exceeds 14 deg, triggering the separation wake (tau = 0) and isotropic 32-deg avalanche slipface.
- Stage 2 (15 - 35 d, Differential Migration & Horn Emergence): Lateral wind deflection (tau_y) and splash diffusion (-D_lat dh/dy) shed sand toward the lower flanks. Because migration speed scales as c(y) ~ 1/H(y), the lower flanks outrun the tall central crest, curving the brink into a concave crescent.
- Stage 3 (35 - 55 d, Equilibrium Barchan Attractor): The dune reaches a steady shape-invariant crescent (H = 3.68 m) with a smooth ~10-deg convex stoss ramp, a 32-deg concave slipface, and symmetric forward horns.

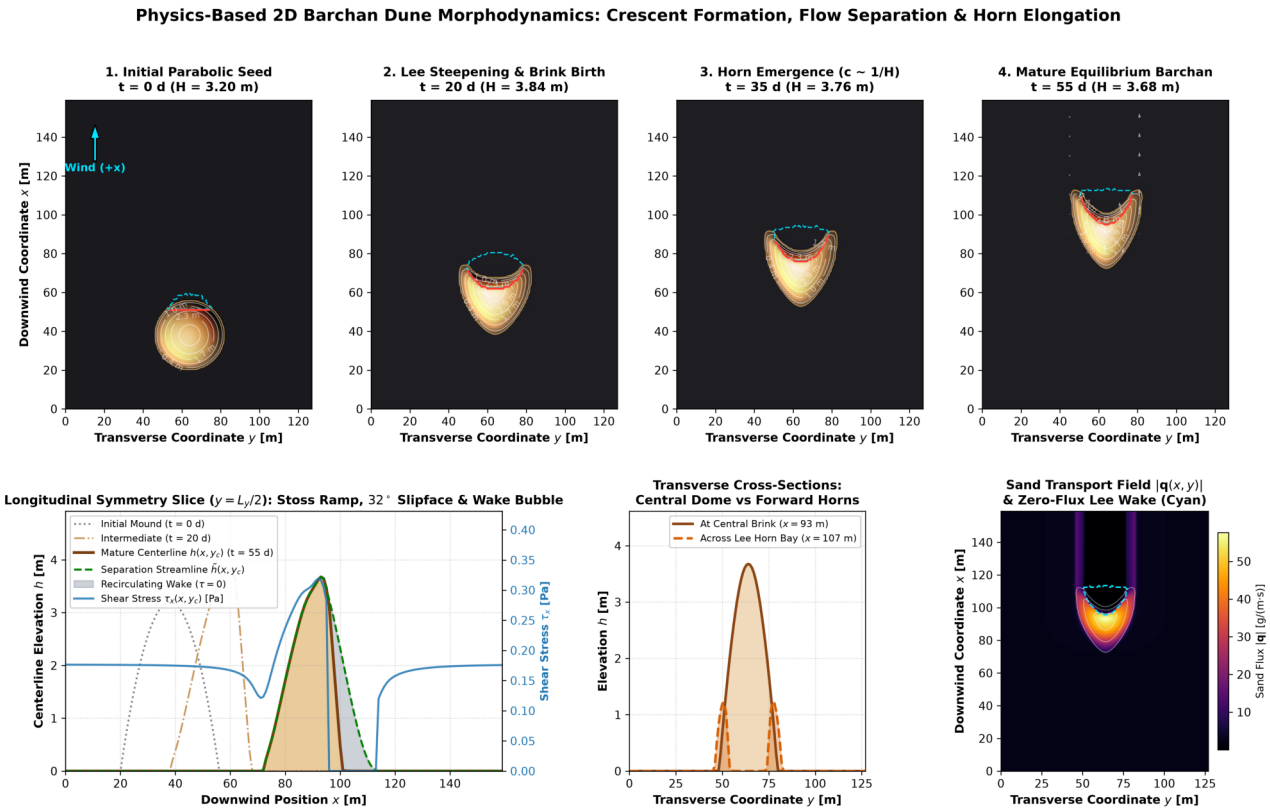


Figure 6 (from 04_barchan_2d_simulation.py): 4-stage planform evolution, centerline profile, horn cross-sections, and sand flux field.

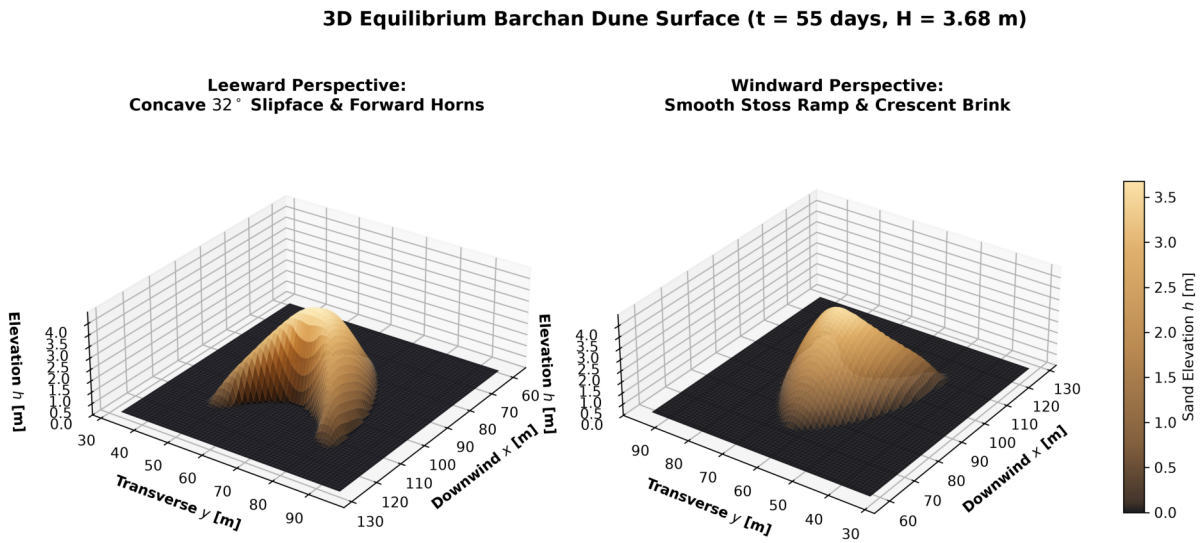


Figure 7 (from 04_barchan_2d_simulation.py): Dual-perspective 3D surface of the emergent equilibrium Barchan dune at t = 55 days.